

Notes: Geng and Pan (WP, 2026)

Model-Based Cross-Section Factors in Corporate Bonds

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1 Big picture

- **Question.** Can one recover economically interpretable common risk factors from the cross-section of U.S. corporate bond returns using a structural asset-pricing model rather than ad hoc bond characteristics?
- **Core answer.** Yes. The paper builds a model with three underlying shocks — credit, duration, and liquidity — and then extracts the associated factor returns from bond-level cross-sectional regressions using *model-implied factor loadings*.
- **Main contribution.** Most factor papers in bonds start from observed characteristics or statistical dimension reduction. This paper instead starts from an arbitrage-free structural model of default with stochastic rates and liquidity shocks, and then uses the model to map each bond into exposures to the three latent risks.
- **Why this matters.** If the loadings are tied directly to economic objects — distance to default, duration, and trading-cost exposure — then the extracted factors should be easier to interpret and more useful for diagnosing market stress.
- **Main pricing result.** The extracted **credit** and **liquidity** factors carry significant risk premia, especially after the global financial crisis, while the **duration** factor has little average premium but still captures major market states and Treasury-market movements.
- **Main asset-pricing result.** The three-factor model helps explain aggregate returns in U.S. corporate bonds, stocks, foreign corporate bonds, and emerging sovereign bonds, and it prices 19 of 20 bond test portfolios sorted on beta, duration, rating, and illiquidity.
- **Main state-diagnostics result.** The factors identify distinct episodes of flight-to-quality, flight-to-liquidity, and flight-to-cash. The paper shows that these are not the same event and can move differently across crises.
- **Main Treasury insight.** Using the duration factor from corporates and Treasuries, the paper decomposes Treasury returns into pure duration risk and a **safety premium**. This lets the authors separate dash-for-cash from flight-to-safety and highlight the unusual Treasury dislocations in March 2020.
- **Main credit-liquidity insight.** Credit shocks predict next-month liquidity shocks strongly, but the reverse relation is much weaker and mostly episodic. So the data support a strong **credit-driven liquidity** channel and only limited **liquidity-driven credit** episodes.
- **Dealer-capacity insight.** The paper links the post-2017 rise in liquidity-factor volatility to reduced dealer net positions in corporate bonds, consistent with tighter balance-sheet capacity after post-crisis regulation.
- **Bottom line.** The paper argues that a structurally grounded factor model can do two things at once: price the cross-section of bonds and open a clearer economic window into how credit, duration, liquidity, and safety interact across markets.

2 Data and measurement (key objects)

2.1 Corporate bond sample

The empirical sample contains fixed-rate, dollar-denominated corporate bonds issued by **non-financial firms** whose equities are listed on NYSE, Nasdaq, or AMEX. The listed-equity restriction is important because the paper needs equity values, equity volatility, and balance-sheet information to estimate firm asset value and default exposure.

The paper excludes bonds with remaining maturity below six months, bonds with issuance size below 100 thousand dollars, firms with too few stock-trading days in a quarter, firms with missing financial statements, and bonds with too few trades to estimate liquidity reliably. To compute the Bao-Pan-Wang gamma illiquidity measure, a bond must trade on at least 11 days in the month.

The final sample runs from **January 2005 to December 2023** and contains about **14,761 bonds** issued by about **1,480 firms**.

Interpretation. The paper studies the tradable, information-rich part of the corporate bond market: large public issuers with both bond and equity data. That gives better measurement of credit exposure, but it also means the sample is tilted toward larger and higher-quality firms.

2.2 Main bond variables

The paper combines data from TRACE, WRDS Bond Returns, Mergent FISD, CRSP, Compustat, and Treasury-rate sources.

Its main bond-side objects are:

- **Bond return:** the month-end excess return from WRDS Bond Returns.
- **Credit spread:** corporate yield minus Treasury yield of the same maturity.
- **Bond beta:** CAPM beta estimated in the bond market over a rolling 24-month window.
- **Maturity:** remaining years to maturity.
- **Rating:** Moody's rating translated into a numerical scale.
- **Age:** years since issuance.
- **Coupon and amount outstanding.**
- **Gamma:** the effective bid-ask-spread proxy of Bao, Pan, and Wang (2011), used as the key illiquidity input.
- **Callable:** indicator for callable bonds.

Interpretation. These are not only descriptive controls. Maturity and gamma feed directly into the paper's factor construction because duration and liquidity exposures come from the structural model, not just from reduced-form sorts.

2.3 Equity and balance-sheet inputs

On the issuer side, the paper uses:

- **Equity size:** market capitalization in logs,

- **Equity beta:** stock CAPM beta from a rolling 60-month window,
- **Book-to-market,**
- **Equity volatility:** annualized volatility from daily stock returns over the previous three months,
- **Leverage:** short- plus long-term debt divided by total assets,
- **Asset growth:** average asset growth over the previous three years.

The default boundary is built from current plus long-term debt, and the effective debt maturity is constructed from debt-category weights using preset maturities.

Interpretation. The paper uses equity-market and balance-sheet information to infer the unobserved firm asset value and asset volatility. That is what allows the model-based credit loading to be tied to distance to default rather than to a reduced-form credit-spread proxy alone.

2.4 Summary statistics and what they imply

The average monthly bond return in the sample is about 0.37%, and the average credit spread is about 1.92%. The median rating is around **Baa1**, the median maturity is around **6.5 years**, and the median amount outstanding is about **500 million dollars**. The sample firms are large: median equity size is roughly **24.5 billion dollars**.

On the factor-loading side, the average liquidity loading is around 0.77, and the average duration loading is around 4.80. The credit loading is often small: the paper notes that about 60% of credit loadings are close to zero because most issuers in the sample are large, relatively safe firms.

Interpretation. This explains an important feature of the results. Because the sample is tilted toward higher-quality issuers, the duration factor naturally matters a lot for safer, longer-maturity bonds, while the credit factor matters most in the lower-quality tail.

2.5 How the liquidity input is measured

The liquidity shock in the structural model is tied to a proportional trading cost κ . Empirically, the paper proxies this cost using the effective bid-ask spread derived from the gamma measure of Bao, Pan, and Wang (2011). In the estimation, it uses the lagged bond-month liquidity measure to map a bond into its loading on the common liquidity factor.

Interpretation. The liquidity factor is not built from raw turnover or Amihud-style price impact. It is built from a model in which a systemic liquidity shock forces an investor to sell at a cost, and gamma is used as the observable counterpart of that cost.

2.6 Monthly and daily factor series

The authors estimate factors at both monthly and daily frequencies. Monthly factors are extracted using month-end bond returns, while daily factors use daily returns with the latest available information from the previous month or prior two quarters for the factor loadings.

Interpretation. Having both frequencies matters. Monthly factors are used for asset-pricing tests and average premia, while daily factors are used to study crisis episodes, rolling correlations, and volatility dynamics.

2.7 Macro and market uncertainty variables

To study time-series drivers of the factors, the paper uses:

- the macro, financial, and real uncertainty indices of Jurado, Ludvigson, and Ng (2015),
- the VIX for equity-market uncertainty,
- the MOVE index for Treasury-rate uncertainty,
- the Hu-Pan-Wang noise measure for Treasury-market liquidity.

Interpretation. These variables are not part of the factor construction. They are used only after the factors are extracted, to test whether the resulting series behave like economically meaningful risk measures.

3 Models and empirical specifications (all equations + interpretation)

3.1 General factor-pricing representation

The paper starts from a generic risk-neutral representation for asset i :

$$\frac{dP_t^i}{P_t^i} - r_t dt = b_i dF_t^Q + dB_t^i, \quad (1)$$

with the risk-neutral factor related to the physical factor by

$$dF_t^Q = \lambda_t dt + dF_t. \quad (2)$$

Interpretation. Bond excess returns can be split into systematic risk exposures (b_i) times factor realizations plus idiosyncratic noise. The paper's entire strategy is to derive the b_i 's from a structural model, then estimate the factor realizations from the cross-section.

3.2 Default risk and interest-rate risk in the structural model

The model combines Merton-style default with Longstaff-Schwartz-style stochastic rates. Aggregate asset value and the short rate evolve as

$$\frac{dV_t}{V_t} = r_t dt + \sigma_V dB_t^V, \quad dr_t = (\alpha - \zeta r_t)dt + \eta dB_t^r. \quad (3)$$

Firm i 's asset value loads on the aggregate credit shock plus idiosyncratic asset noise. The payoff to risky debt is the minimum of firm value and promised payment, so the bond price becomes a function of firm asset value, the short rate, and time to maturity.

Interpretation. This is the core economic structure. Credit risk comes from how close firm value is to the default boundary; duration risk comes from sensitivity to the short-rate process.

3.3 Closed-form risky bond price

Under the model, the risky zero-coupon bond price can be written in closed form as

$$P_t^i = KR(r_t, t, T)N(d_2) + V_t^i R(r_t, t, T)e^{M(r_t, t, T) + \frac{1}{2}S^2(t, T)}N(-d_1). \quad (4)$$

The exact formulas for d_1 , d_2 , M , and S come from the Longstaff-Schwartz extension of Merton.

Interpretation. The first term is the discounted promised payment weighted by survival/default probabilities, while the second term captures the state-contingent asset-value piece. Because this price is explicit, the paper can differentiate it with respect to economic state variables and obtain factor loadings analytically.

3.4 Model-implied factor loadings

The structural loadings are the sensitivities of log bond price to the model's state variables:

$$b_{i,t}^V = \frac{\partial \ln P_t^i}{\partial \ln V_t}, \quad b_{i,t}^r = \frac{\partial \ln P_t^i}{\partial r_t}. \quad (5)$$

In words, b^V is the **credit** loading and b^r is the **duration** loading. Economically, the credit loading is closely related to distance to default, and the duration loading is essentially a modified-duration-type interest-rate exposure.

Interpretation. The paper's credit factor is not just "high spread minus low spread." It is the factor that prices securities in proportion to how sensitive they are to systematic shifts in firm asset value. Likewise, the duration factor prices securities in proportion to how sensitive they are to the short rate.

3.5 Adding liquidity shocks

The paper augments the model with a Poisson liquidity shock. When the shock arrives, the investor must sell at a proportional cost κ_i . The bond return equation becomes

$$\frac{dP_t^i}{P_t^i} - r_t dt = b_{i,t}^V dF_t^V + b_{i,t}^r dF_t^r + b_{i,t}^N dF_t^N + dB_t^i, \quad (6)$$

where

$$b_{i,t}^N = e^{-\kappa_i} - 1. \quad (7)$$

Empirically the paper uses the sign-reversed version of the rate and liquidity loadings so that larger values correspond more naturally to larger duration and liquidity exposure.

Interpretation. The liquidity factor is modeled as a *systemic forced-sale shock*. Bonds with larger trading costs lose more when the shock hits, so they load more on common liquidity risk.

3.6 KMV-style inversion for firm asset value

To compute the model-implied credit loading, the paper must infer the firm's asset value V_t^i and asset volatility σ from observed equity value and equity volatility. It follows the KMV logic and solves

$$E_t^i = E_t^i(V_t^i, \sigma), \quad \sigma_E = \frac{V_t^i}{E_t^i} \frac{\partial E_t^i}{\partial V_t^i} \sigma, \quad (8)$$

where E_t^i is equity value and σ_E is equity volatility.

Interpretation. Equity is treated as a contingent claim on firm assets. By matching both equity value and equity volatility, the paper backs out the unobserved firm asset value and asset volatility needed for credit-risk measurement.

3.7 Cross-sectional regression used to extract factor returns

Given the bond-level loadings, the paper extracts factor realizations month by month (or day by day) from the cross-section using a regression of bond excess returns on the three model-implied loadings:

$$R_{i,t}^e = \lambda_t^V b_{i,t}^V + \lambda_t^r b_{i,t}^r + \lambda_t^N b_{i,t}^N + \varepsilon_{i,t}. \quad (9)$$

The time- t slope coefficients are the time- t realizations of the **credit**, **duration**, and **liquidity** factors. Stacking these slopes over time yields the factor time series.

Interpretation. This is the paper's key empirical move. The factors are not portfolio returns chosen in advance; they are the cross-sectional prices of economically derived exposures.

3.8 Market-level pricing regression

To test whether the three factors price broader markets, the paper estimates

$$R_t^M - R_t^f = \alpha + \beta^V F_t^V + \beta^N F_t^N + \beta^r F_t^r + \varepsilon_t, \quad (10)$$

for market returns in U.S. corporate bonds, high-yield bonds, investment-grade bonds, stocks, Treasuries, foreign corporate bonds, and sovereign debt portfolios.

Interpretation. If the factors extracted from the corporate bond cross-section are truly systematic, they should help explain returns outside the exact bonds used in the extraction. This is the paper's out-of-sample economic validation.

3.9 Treasury decomposition and predictive regressions

A central application defines

$$SafetyPremium_t = Duration_t^{UST} - Duration_t^{Corp}, \quad (11)$$

where $Duration^{Corp}$ is the duration factor from investment-grade corporates and $Duration^{UST}$ is the duration factor from Treasuries.

The paper also studies lead-lag relations using regressions of the form

$$Liquidity_{t+1} = a + b Credit_t + u_{t+1}, \quad Credit_{t+1} = a + c Liquidity_t + v_{t+1}. \quad (12)$$

Interpretation. The first equation asks whether worsening credit conditions forecast future liquidity deterioration. The second asks whether illiquidity today predicts future credit stress. The asymmetry between b and c is one of the paper’s central findings.

4 Factor construction and identification (what makes the approach work)

4.1 What identifies each factor

The paper’s three-factor extraction works because different bonds load differently on the three model states:

- bonds issued by safer versus riskier firms differ in **credit** exposure,
- short- versus long-maturity bonds differ in **duration** exposure,
- more versus less costly-to-trade bonds differ in **liquidity** exposure.

Once the model maps each bond into those three exposures, the cross-section at time t identifies the factor realizations from the slope coefficients.

Interpretation. Identification is cross-sectional. The model does not need one bond to reveal all three factors. It needs a large panel with enough variation in default distance, maturity, and liquidity to separate the common shocks.

4.2 Why model-based loadings are different from characteristic-based factors

The paper emphasizes that its factors are not constructed from raw bond characteristics such as rating buckets, maturity buckets, or turnover sorts. Instead, the characteristics feed into a structural pricing model, and the extracted factor loadings are the model’s derivatives of log price with respect to economic state variables.

Interpretation. This is the paper’s conceptual pitch: a maturity variable by itself is not the duration factor, and a rating variable by itself is not the credit factor. The factor should come from the pricing kernel of the model, not from a sorting rule.

4.3 Where the clean separation comes from

The model-based approach helps the paper separate risks that often move together in practice. A bond can be long duration but safe, or risky but short maturity, or safe but illiquid. Because the loadings are estimated bond by bond, the cross-section can use these mixed combinations to disentangle common credit, duration, and liquidity shocks.

Interpretation. This is why the paper can later distinguish flight-to-quality from flight-to-cash and show that the two need not coincide.

4.4 Empirical assumptions behind the construction

The factor construction relies on several important assumptions:

- the structural model is a reasonable approximation to bond pricing,
- the KMV inversion recovers meaningful firm asset values and volatilities,
- gamma is an adequate proxy for bond-level trading-cost exposure,
- the listed-equity sample contains enough cross-sectional variation to identify the factor slopes,
- lagged balance-sheet and liquidity inputs are measured with manageable error.

Interpretation. The factors are only as good as the model and its measurement inputs. The approach is more structured than a pure statistical factor model, but it is also more assumption-intensive.

4.5 What the paper cannot fully resolve

The sample is concentrated in larger, publicly listed issuers, so the extracted factors may underrepresent the very distressed tail of the corporate bond market. The credit loading is small for many high-quality firms, which likely contributes to the residual alpha in the highest-rated portfolio. The paper itself points to the credit spread puzzle and notes that an additional crash-risk ingredient may still matter for the safest bonds.

Interpretation. The model performs well, but it is not the last word. Its strongest success is in cleanly organizing a large portion of cross-sectional and time-series variation, not in claiming that all bond returns are fully explained.

5 Tables and figures

5.1 Table I and Table II: sample structure and factor premia

Read: Table I shows a large, liquid, mostly investment-grade sample: roughly 14,761 bonds from 1,480 firms over 2005–2023, average monthly bond return around 0.37%, average credit spread around 1.92%, median rating around Baa1, and median maturity around 6.5 years. Table II then reports the factor premia. Monthly **credit** premium is about 24 bps in the full sample and 39 bps after 2009; monthly **liquidity** premium is about 4 to 5 bps and strongly significant; monthly **duration** premium is small and statistically weak.

Economic message. The first big message is that the paper’s factors are not statistical artifacts. At least two of them — credit and liquidity — earn economically meaningful average premia, and the premia become stronger after the crisis.

Table I
Summary Statistics

This table reports summary statistics for our sample of bonds. The sample period is from January 2005 to December 2023. NumBonds is the number of bonds. BondRet is the month-end bond return from WRDS Bond Returns database. CreditSpread is the difference in yield between corporate bonds and Treasury bonds of the same maturity. BondBeta is the bond beta estimated from a CAPM model in the bond market over a rolling window of past 24 months. Maturity is the bond's time to maturity in years. Rating is a numerical translation of Moody's rating: 1=Aaa and 21=C. Age is the time since issuance in years. Coupon is the bond's coupon payment in percent. Amount is the bond's amount outstanding in millions of dollars. Gamma is the illiquidity measure from Bao, Pan, and Wang (2011). Callable is 1 for bonds issued with callable options. NumIssuers is the number of issuers. EquitySize is the log of the firm's equity size. EquityBeta is the equity beta estimated from a CAPM model in the stock market over a rolling window of past 60 months. BM is the book value of the equity to market capitalization. EquityVolatility is the annualized equity volatility. Leverage is the ratio of total short- and long-term debt to the total asset value. AssetGrowth is the average growth rate of the asset value in the past three years. b_C^V , b_L^V , and b_D^V are the bond's factor loading associated with credit, duration, and liquidity factors, respectively.

Variable	Mean	StdDev	Min	P25	Median	P75	Max
Bond-Level Variables							
NumBonds	14,761						
BondRet (%)	0.37	2.68	-9.25	-0.53	0.27	1.31	10.23
CreditSpread (%)	1.92	2.13	-0.09	0.75	1.28	2.22	14.11
BondBeta	1.12	0.87	-0.44	0.50	0.95	1.57	5.53
Maturity (yr)	9.47	8.53	0.50	3.26	6.13	14.05	29.99
Rating	8.59	3.33	1.00	6.00	8.00	10.00	17.00
Age (yr)	5.67	4.73	1.00	2.00	4.00	7.00	24.00
Coupon (%)	5.13	1.90	1.20	3.65	5.10	6.50	9.75
Amount	659	520	1.80	300	500	800	2850
Gamma (bps)	0.83	2.20	-1.09	0.03	0.17	0.65	16.76
Call	0.88	0.32	0.00	1.00	1.00	1.00	1.00
Equity-Level Variables							
NumIssuers	1,480						
EquitySize (log)	23.79	1.62	18.46	22.87	23.92	24.85	27.84
EquityBeta	1.02	0.62	-0.08	0.56	0.95	1.34	3.50
BM	0.46	0.40	-0.78	0.22	0.41	0.64	2.23
EquityVolatility	0.30	0.19	0.09	0.18	0.25	0.35	1.29
Leverage	0.39	0.17	0.07	0.27	0.37	0.48	0.97
AssetGrowth	0.07	0.11	-0.17	0.01	0.05	0.10	0.53
Estimated Factor Loadings							
b_C^V	0.10	0.29	0.00	0.00	0.00	0.02	2.09
b_L^V	5.87	4.30	-1.94	2.64	4.80	7.79	16.70
b_D^V	1.19	1.35	0.00	0.31	0.77	1.57	8.11

Table II
Factor Returns

This table reports the average returns and CAPM alphas for monthly factors and daily factors. Reported in square brackets are *t*-statistics. *, **, and *** indicate significant at the 10%, 5%, and 1% level, respectively.

Average Factor Returns and Alphas						
	Monthly (%)				Daily (bps)	
	2005-2023		2009-2023		2005-2023	
	Mean	Alpha	Mean	Alpha	Mean	Alpha
Credit	0.24*	0.06	0.39***	0.13	0.82	-0.61
	[1.90]	[0.61]	[2.86]	[1.21]	[1.32]	[-1.07]
Liquidity	0.04***	0.04***	0.05***	0.05***	0.34**	0.32**
	[4.29]	[3.98]	[4.45]	[3.98]	[2.39]	[2.23]
Duration	0.01	-0.02*	0.02	-0.02**	0.12	-0.18***
	[0.83]	[-1.78]	[1.08]	[-2.20]	[1.53]	[-3.17]
					[1.90]	[-4.36]

5.2 Figures 1–3 and Table III: factor dynamics, crisis episodes, and macro drivers

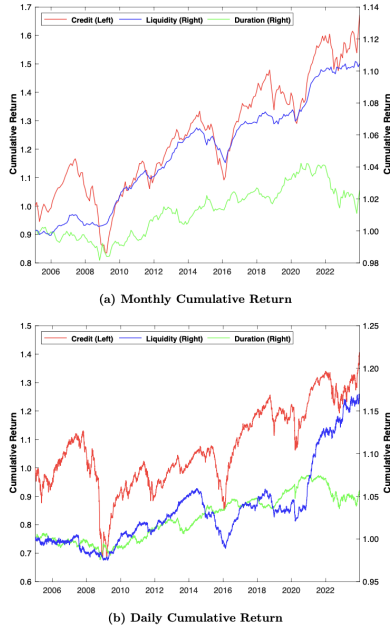


Figure 1. This figure plots the cumulative return of our three factors – credit, liquidity, and duration – at a monthly frequency in Panel (a) and a daily frequency in Panel (b).

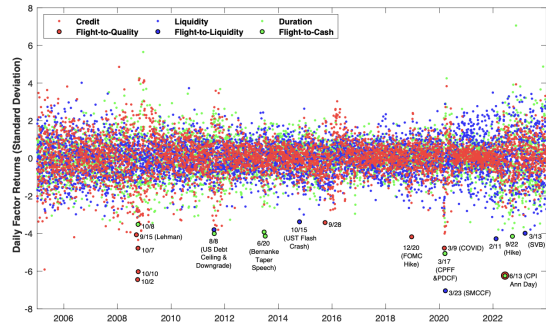


Figure 2. Factor Normalized Returns. This figure plots the normalized daily returns of our three factors: credit, liquidity and duration.

Read: Figure 1 plots cumulative returns of the monthly and daily factors. Credit and liquidity trend upward on average but collapse in stress periods, while duration is less remunerative on average and tracks large rate regimes. Figure 2 marks extreme daily moves and identifies episodes

of flight-to-quality, flight-to-liquidity, and flight-to-cash. Figure 3 shows that the first principal component explains about 60% of total factor variation, with the second and third components explaining about 25% and 15%, respectively. The Shapley decomposition shows credit dominating during the 2008 crisis and the 2016 junk-bond episode, liquidity becoming especially important after 2017, and duration surging in importance during the 2022 tightening cycle. Table III links the extracted factors to macro drivers: credit loads most negatively on VIX and financial uncertainty, liquidity is especially tied to financial uncertainty, and duration is most strongly associated with MOVE.

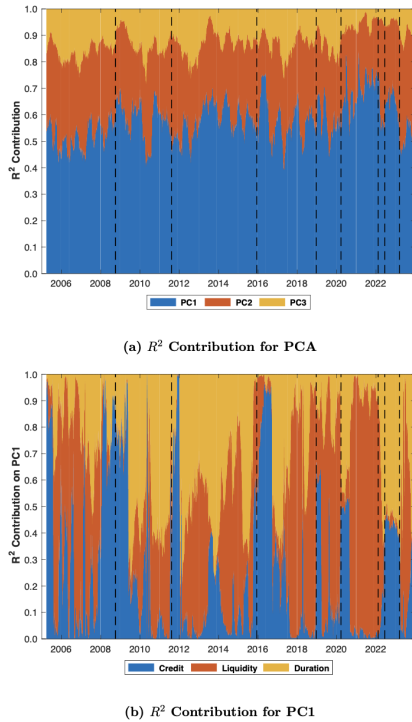


Figure 3. This figure plots the R^2 Contribution from PCA analysis based on our three factors daily returns, credit, liquidity and duration, over a rolling window of past three months. Panel (a) shows the R^2 contribution of each three PC in explaining the three factor returns. Panel (b) shows the Shapley R^2 decomposition of PC1 with respect to three factors.

Table III
Macro Drivers of Three Factors (Num of STDs)

This table reports the result of time-series regressions of factor returns on a set of macro and market uncertainty variables, including the macro uncertainty index, financial uncertainty index, and real uncertainty index developed by Jurado, Ludvigson, and Ng (2015), the VIX index from the CBOE, the MOVE index, and the noise measure from Hu, Pan, and Wang (2013). Reported in square brackets are t -statistics. *, **, and *** indicate significant at the 10%, 5%, and 1% level, respectively.

	Credit		Liquidity		Duration	
Fin Uncertainty	-0.35*** [-4.14]	-0.18** [-2.19]	-0.25*** [-3.31]	-0.25*** [-3.05]	-0.17** [-2.02]	-0.05 [-0.63]
Macro Uncertainty	-0.17 [-1.06]	-0.10 [-0.71]	-0.34** [-2.11]	-0.35** [-2.11]	-0.01 [-0.08]	0.05 [0.30]
Real Uncertainty	0.02 [0.16]	-0.03 [-0.19]	0.19 [1.26]	0.20 [1.37]	-0.00 [-0.03]	-0.05 [-0.35]
VIX	-0.42*** [-5.60]	-0.32*** [-4.19]	-0.15 [-1.62]	-0.02 [-0.25]	-0.18** [-2.13]	-0.16* [-1.85]
MOVE	-0.11* [-1.74]	-0.10* [-1.73]	0.05 [0.56]	0.07 [0.97]	-0.19** [-2.35]	-0.19** [-2.32]
Noise	-0.15** [-2.31]	-0.08 [-1.24]	-0.11 [-1.49]	0.00 [0.03]	-0.04 [-0.53]	-0.04 [-0.44]
Constant	-0.00 [-0.00]	0.00 [0.08]	0.00 [0.07]	0.00 [0.00]	-0.00 [-0.07]	-0.00 [-0.07]
Obs	228	227	227	228	227	228
R ²	0.20	0.28	0.33	0.15	0.03	0.14
				0.02	0.02	0.09

Economic message. These exhibits do two jobs. First, they validate the factors as economically recognizable state variables. Second, they show that the dominant source of bond-market risk is not constant over time: it rotates across credit, liquidity, and duration depending on the regime.

5.3 Table IV: explaining aggregate market returns across asset classes

Read: Table IV regresses aggregate excess returns on the extracted factors. Average excess returns are economically large in U.S. corporate bonds, high-yield bonds, investment-grade bonds, stocks, foreign corporate bonds, and emerging sovereign bonds. With only the credit factor, much of the alpha in HY, IG, stocks, and emerging-market sovereign bonds already shrinks markedly. In the full three-factor model, explanatory power improves further, especially for investment-grade bonds and foreign corporate bonds. Investment-grade bonds load strongly on duration and positively on liquidity, while Treasuries load negatively on credit in the one-factor specification and show a

positive residual in the full model.

Economic message. The paper’s factors come from corporate bonds, but they travel well. Credit is the main systematic force across risky assets, yet duration and liquidity add economically relevant dimensions, especially for safer fixed-income markets.

Table IV
Explaining the Market Excess Returns

This table reports the result of monthly panel regressions of market-level excess returns on our three model-based factors: credit, liquidity and duration. We consider five market indices in US, including CorpBond (Aggregate Corporate Bond Market Index), HY (Bloomberg Barclays High-Yield Index), IG (Bloomberg Barclays Investment-Grade Index), Stock (Fama-French MktRF Portfolio), and Treasury (Bloomberg Barclays Aggregate Treasury Index), as well as three market indices for Foreign countries, including CorpBond (aggregate corporate dollar bond index by foreign issuers), SovereignEM (Bloomberg Emerging Market Sovereign Index), and SovereignEU (Bloomberg EU Sovereign Index). Panel A reports the mean excess returns. Panel B reports the regression results after controlling for credit factor. Panel C reports the regression results after controlling for our three factors. Reported in square brackets are t -statistics. *, **, and *** indicate significant at the 10%, 5%, and 1% level, respectively.

Panel A: Mean Excess Return								
	US					Foreign		
	CorpBond	HY	IG	Stock	Treasury	CorpBond	SovereignEM	SovereignEU
Mean	0.30** [2.53]	0.49** [2.47]	0.26* [1.95]	0.76** [2.55]	0.13 [1.45]	0.34*** [2.69]	0.36** [2.16]	0.13 [1.22]
Panel B: Credit Factor Model								
α	0.16 [1.49]	0.19 [1.43]	0.14 [1.13]	0.34 [1.63]	0.16* [1.72]	0.19* [1.74]	0.18 [1.11]	0.04 [0.39]
Credit	0.59*** [8.20]	1.25*** [10.60]	0.47*** [5.25]	1.77*** [14.85]	-0.13** [-2.05]	0.64*** [7.72]	0.78*** [5.58]	0.36*** [4.13]
Obs	228	228	228	228	228	228	228	228
R^2	0.34	0.63	0.20	0.55	0.03	0.41	0.34	0.18
Panel C: Three Factors Model								
α	0.09 [1.20]	0.16 [1.17]	0.06 [0.77]	0.35 [1.62]	0.13** [2.12]	0.07 [0.81]	0.08 [0.55]	-0.02 [-0.25]
Duration	6.20*** [10.37]	1.86 [1.52]	7.44*** [14.23]	0.88 [0.64]	5.49*** [16.36]	5.67*** [10.68]	5.76*** [6.52]	4.39*** [7.33]
Credit	0.29*** [4.48]	1.16*** [7.06]	0.12* [1.95]	1.75*** [11.39]	-0.36*** [-9.36]	0.32*** [4.94]	0.47*** [3.43]	0.13* [1.66]
Liquidity	1.58*** [2.07]	0.62 [0.39]	1.69** [2.12]	-0.37 [-0.19]	0.53 [0.90]	2.78*** [3.06]	2.32* [1.82]	1.48* [1.69]
Obs	228	228	228	228	228	228	228	228
R^2	0.76	0.64	0.77	0.55	0.71	0.73	0.53	0.48

Table V
Explaining the Bond Market Testing Portfolios

This table reports the result of monthly panel regressions of bond market testing portfolios on our three model-based factors: credit, liquidity and duration. We construct four categories of quintile portfolios based on bond CAPM beta (Panel A), bond duration (Panel B), bond rating (Panel C), and illiquidity (Panel D). Reported in square brackets are t -statistics. *, **, and *** indicate significant at the 10%, 5%, and 1% level, respectively.

	Panel A: BondBeta Sorted Portfolio					Panel B: Duration Sorted Portfolio				
	G1	G2	G3	G4	G5	G1	G2	G3	G4	G5
Mean	Raw Excess Monthly Return					Raw Excess Monthly Return				
	0.13** [2.01]	0.20** [2.35]	0.28** [2.26]	0.33** [2.03]	0.46** [2.11]	0.17*** [2.66]	0.25** [2.37]	0.30** [2.19]	0.37** [2.25]	0.34 [1.59]
α	Three Factors Model					Three Factors Model				
	0.01 [0.17]	0.06 [0.85]	0.09 [1.02]	0.06 [0.63]	0.11 [1.09]	0.05 [0.90]	0.05 [0.62]	0.08 [0.75]	0.09 [0.88]	0.07 [0.97]
Duration	1.95*** [4.43]	3.75*** [7.10]	5.98*** [7.65]	8.40*** [10.56]	9.77*** [8.47]	1.09*** [2.80]	2.62*** [4.65]	4.43*** [5.92]	8.45*** [11.30]	13.41*** [17.17]
Credit	0.18*** [3.22]	0.13** [2.05]	0.20** [2.44]	0.32*** [3.94]	0.73*** [4.57]	0.24*** [4.49]	0.44*** [5.45]	0.50*** [4.90]	0.30*** [3.95]	0.19*** [2.74]
Liquidity	1.20** [1.98]	1.42** [2.05]	1.58* [1.78]	1.98* [1.86]	1.47 [1.03]	1.04* [1.88]	1.42* [1.79]	1.05 [1.07]	2.29** [2.14]	1.50* [1.86]
R^2	0.48	0.57	0.65	0.77	0.80	0.46	0.60	0.62	0.77	0.92
Mean	Panel C: Rating Sorted Portfolio					Panel D: Illiquidity Sorted Portfolio				
	Raw Excess Monthly Return					Raw Excess Monthly Return				
	0.22* [1.88]	0.25* [1.95]	0.27** [1.98]	0.34** [2.19]	0.39* [1.92]	0.25* [1.81]	0.23** [2.47]	0.27** [2.24]	0.36** [2.31]	0.39** [2.06]
α	Three Factors Model					Three Factors Model				
	0.11** [2.39]	0.08 [1.26]	0.05 [0.66]	0.05 [0.44]	0.02 [0.12]	0.07 [0.73]	0.10 [1.64]	0.09 [1.19]	0.09 [1.11]	-0.05 [-0.47]
Duration	7.62*** [15.45]	7.67*** [14.43]	7.48*** [12.41]	5.26*** [5.46]	1.85* [1.92]	5.31*** [6.11]	4.57*** [9.97]	6.07*** [10.88]	7.91*** [11.57]	8.43*** [10.88]
Credit	-0.06 [-1.32]	0.06 [1.18]	0.02 [0.33]	0.55*** [4.48]	1.22*** [8.40]	0.42*** [3.98]	0.16*** [2.62]	0.23*** [3.75]	0.35*** [4.85]	0.58*** [6.91]
Liquidity	0.70 [1.44]	1.22* [1.86]	1.77** [2.16]	2.10 [1.62]	1.32 [0.99]	0.16 [0.15]	0.76 [1.32]	1.13 [1.55]	1.86** [2.11]	4.63*** [3.84]
R^2	0.85	0.82	0.79	0.65	0.71	0.64	0.68	0.76	0.80	0.81

5.4 Table V: explaining the bond test portfolios

Read: Table V constructs 20 bond portfolios sorted by bond beta, duration, rating, and illiquidity. Raw excess returns range from roughly 13 to 46 bps per month. After controlling for the three factors, 19 of 20 alphas become insignificant. The pattern of loadings is intuitive: credit loadings rise sharply with bond beta and with worse ratings; duration loadings rise strongly with portfolio duration and are strongest for safer bonds; liquidity loadings become much larger for less liquid portfolios.

Economic message. This is the paper’s cleanest cross-sectional pricing success. The factors do not merely explain aggregate market indices; they organize the cross-section in exactly the way the economic interpretation predicts.

5.5 Table VI and Figures 4–6: Treasury decomposition and the credit-spread/Treasury-yield relation

Read: Table VI and Figures 4–6 are the conceptual center of the paper. The paper defines a **safety premium** as the Treasury duration factor minus the corporate duration factor. Figure 4 shows that this return-based safety-premium measure comoves strongly with the Treasury-basis convenience-yield proxy from Jiang, Krishnamurthy, and Lustig, and the relation becomes stronger in BondSafety months. Figure 5 then places each day in a plane whose x-axis is pure duration risk and y-axis is the safety premium, allowing the paper to separate ordinary flight-to-safety from

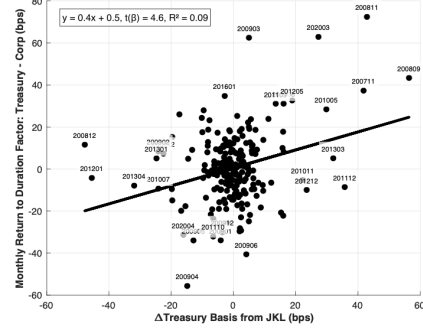
Table VI
Decompose UST Returns

This table reports time-series regressions of monthly excess returns for four asset classes on the two components of Treasury duration factor (*DurationUST*): the pure duration risk factor (*DurationCorp*) and the safety premium (*SafetyPremium*). Panels A through D present results for (1) aggregate equity market, (2) U.S. Treasuries, (3) investment-grade (IG) corporate bonds, and (4) high-yield (HY) corporate bonds. Panels E and F examine the relationships with credit and liquidity factors, respectively. Reported in square brackets are t-statistics. *, **, and *** indicate significant at the 10%, 5%, and 1% level, respectively.

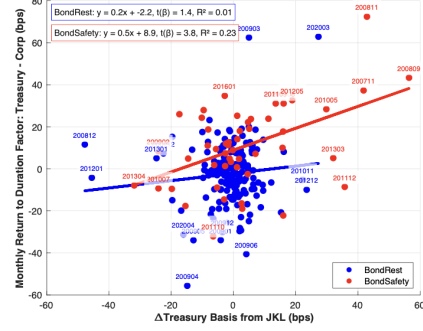
	Panel A: Stock Excess Return				Panel B: Treasury Excess Return			
DurationUST	-0.02 [-1.28]				0.05*** [32.28]			
DurationCorp	0.06*** [3.84]		0.03** [2.57]		0.04*** [13.67]		0.05*** [28.61]	
SafetyPremium			-0.17*** [-6.59]				0.04*** [6.45]	
Constant	0.80*** [2.65]	0.68** [2.36]	0.85*** [3.57]	0.79*** [3.33]	0.04 [1.13]	0.07 [1.14]	0.11 [1.39]	0.04 [1.14]
Obs	228	228	228	228	228	228	228	228
R ²	0.01	0.09	0.36	0.39	0.89	0.50	0.19	0.89

	Panel C: IG Excess Return				Panel D: HY Excess Return			
DurationUST	0.05*** [7.59]				-0.01 [-0.67]			
DurationCorp	0.07*** [18.47]		0.07*** [18.77]		0.05*** [4.38]		0.03*** [3.53]	
SafetyPremium			-0.04*** [-3.05]				-0.12*** [-7.30]	
Constant	0.17 [1.58]	0.15** [2.24]	0.28** [2.27]	0.17** [2.60]	0.51** [2.53]	0.42** [2.24]	0.55*** [3.88]	0.50*** [3.63]
Obs	228	228	228	228	228	228	228	228
R ²	0.33	0.74	0.11	0.76	0.01	0.16	0.46	0.52

	Panel E: Credit Factor				Panel F: Liquidity Factor			
DurationUST	-0.01* [-1.69]				-0.25*** [-5.86]			
DurationCorp	0.03*** [5.01]		0.02*** [3.95]		-0.16*** [-3.62]		-0.22*** [-4.75]	
SafetyPremium			-0.08*** [-11.51]				-0.27*** [-4.17]	
Constant	0.26** [2.08]	0.20* [1.69]	0.28*** [3.15]	0.25*** [2.97]	4.80*** [5.24]	4.55*** [4.62]	4.46*** [4.62]	4.79*** [5.27]
Obs	228	228	228	228	228	228	228	228
R ²	0.02	0.12	0.50	0.53	0.17	0.06	0.08	0.18



(a) Treasury Basis Comparison



(b) Treasury Basis Comparison by Group

Figure 4. This figure plots the change in the Treasury basis from Jiang, Krishnamurthy, and Lustig (2021) against our safety premium measure, defined as the monthly return difference in duration factors estimated from the U.S. Treasury and corporate-bond markets, respectively. Panel A uses the full sample; Panel B uses the subsample. BondSafety denotes months with pronounced flight-to-safety episodes – sharp equity declines coupled with 10-year Treasury rallies (See Hu, Jin, and Pan (2025)).

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dash-for-cash and, more unusually, from flight-away-from-safety. The March 2020 points show how special the COVID Treasury dislocation was: on dates such as 2020-03-10 and 2020-03-17, investors simultaneously ran from duration and from Treasury safety. Table VI also shows that equities and HY bonds load positively on pure duration risk and negatively on the safety premium, while Treasury returns load positively on both. Finally, Figure 6 shows that the famous negative correlation between credit spreads and Treasury yields is not mainly a pure duration story; it is driven largely by the safety-premium and liquidity channels.

Economic message. The paper turns an old stylized fact into a more precise statement. “Treasuries rally when credit worsens” is too coarse. Sometimes the relevant force is duration, sometimes safety, sometimes liquidity, and in March 2020 even Treasuries temporarily lost their safe-haven role.

5.6 Figures 7 and 8: lead-lag asymmetry and dealer balance-sheet capacity

Read: Figure 7 studies predictive relations between credit and liquidity. Current credit strongly predicts next-month liquidity: the full-sample coefficient is about 0.05 with a t-stat around 8.56 and predictive R^2 around 25%. The reverse relation is much weaker: current liquidity predicts next-month credit only weakly on average, with significance concentrated around the aftermath of 2008 and parts of the 2022 tightening cycle. Figure 8 studies volatility of the factor returns and links liquidity volatility to dealer inventories. Liquidity-factor volatility rises substantially after

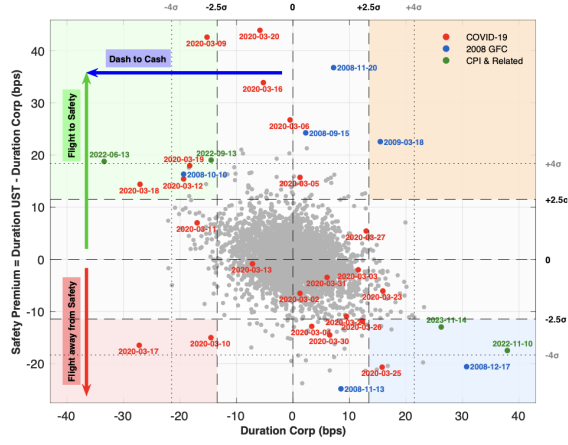


Figure 5. Flight-to-Safety vs Dash-to-Cash. This figure plots the daily duration risk factor (*DurationCorp*) against the safety premium component (*SafetyPremium*) based on our decomposition of U.S. Treasury returns. The red points highlight March 2020 during the COVID-19 crisis, blue points mark key dates from the 2008 financial crisis, green points indicate CPI announcement days and related events, and gray points represent all other trading days. The colored regions delineate different market regimes: dash-for-cash (blue arrow), flight-to-safety (green arrow), and flight-away-from-safety (red arrow). Reference lines at $\pm 2.5\sigma$ and $\pm 4\sigma$ demarcate extreme observations in both dimensions.

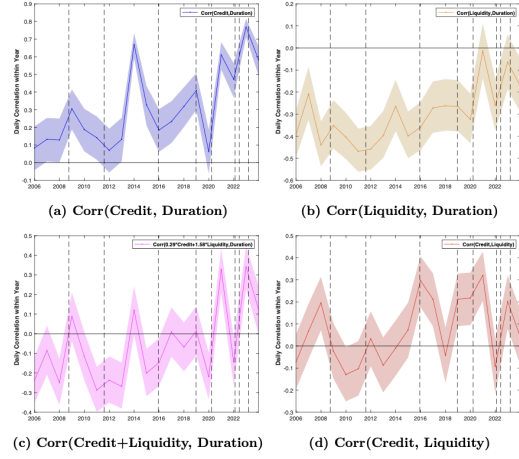


Figure 6. This figure plots the time-varying annual correlations among credit risk, liquidity risk, and duration risk. Panel (a) shows the correlation between credit and duration factors; Panel (b) displays the correlation between liquidity and duration factors; Panel (c) illustrates the correlation between duration and a combined credit-liquidity measure, where the weights are estimated from regressing corporate bond returns on the three factors; Panel (d) exhibits the correlation between credit and liquidity factors directly. All correlations are computed using daily factor returns within each calendar year. The shaded bands represent 95% confidence intervals.

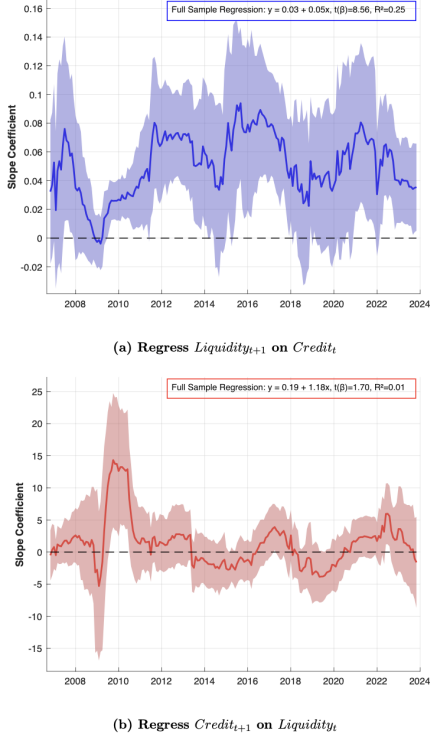


Figure 7. This figure plots time-varying lead-lag relationships between credit and liquidity risk factors. Panel (a) displays rolling window estimates from regressing the liquidity factor at month $t+1$ on the credit factor at month t . Panel (b) shows the reverse predictive relationship. Each regression is estimated using a 24-month rolling window ending at time t , with the resulting slope coefficient plotted at the end date of each estimation window. The shaded areas represent 95% confidence intervals. Full-sample regression results are reported in the text boxes.

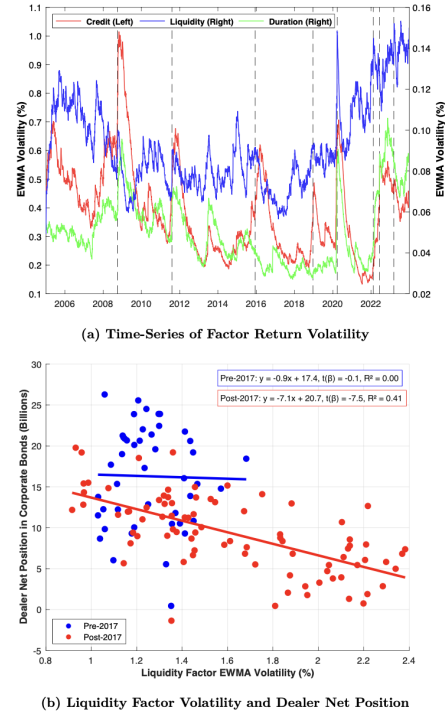


Figure 8. Panel (a) plots the EWMA volatility of our three factors, credit, liquidity and duration, with decay parameter 0.98. Panel (b) shows a scatter plot of the EWMA volatility of the liquidity factor against primary dealers' net positions in investment-grade and high-yield corporate bonds, estimated using data from the Federal Reserve Bank of New York's website.

2017, and from 2017 to 2023 it is strongly negatively associated with primary dealers' net positions in corporate bonds, whereas the relation is statistically weak before 2017.

Economic message. These are the paper's sharpest mechanism results. First, credit deterioration tends to create future illiquidity more reliably than illiquidity creates future credit deterioration. Second, post-crisis dealer constraints seem to have made market liquidity more fragile, and the paper's liquidity factor captures that fragility.

6 Short summary

- The paper extracts three common risk factors from corporate bond returns: credit, duration, and liquidity.
- The novelty is that the factors come from *model-implied factor loadings* rather than from simple bond characteristics or purely statistical factors.
- The structural model combines Merton-style default, Vasicek interest rates, and a Poisson liquidity shock.
- Firm asset values and volatilities are inferred using a KMV-style inversion from equity values and equity volatility.
- Cross-sectional regressions of bond returns on the three loadings deliver monthly and daily factor returns.
- Credit and liquidity factors earn significant risk premia; duration has weak average premium but large time-series importance.
- The three factors explain aggregate returns across several asset classes and price 19 of 20 bond test portfolios.
- A key application decomposes Treasury returns into pure duration risk and a safety premium, helping distinguish flight-to-safety from dash-for-cash.
- The negative relation between credit spreads and Treasury yields is shown to run mainly through safety-premium and liquidity channels.
- Credit predicts future liquidity strongly, while liquidity predicts future credit only episodically.
- After 2017, liquidity-factor volatility rises and is tightly linked to dealer balance-sheet capacity.
- Overall, the paper argues that a structurally grounded three-factor framework can both price bond returns and diagnose the economic source of major market events.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that one can recover economically interpretable common factors from the cross-section of corporate bond returns by starting from a structural model of bond pricing and using the resulting model-implied loadings to estimate the factor realizations.

7.2 Economic intuition

The deepest idea is simple. A bond's price reacts to three broad state variables: the underlying health of the issuer, the discount-rate environment, and market-wide liquidity conditions. If a model tells us how sensitive each bond is to each state variable, then the cross-section of returns on many different bonds reveals the common shocks hitting those states.

7.3 What the paper adds

The paper adds both a method and a set of applications. Methodologically, it offers a model-based alternative to characteristic-sorted or purely statistical bond factors. Empirically, it shows that the extracted factors help explain asset returns across markets and clarify several classic fixed-income facts: Treasury convenience yield, the credit-spread/Treasury-yield relation, the asymmetry between credit-driven and liquidity-driven stress, and the role of dealer balance-sheet capacity in post-crisis market liquidity.

7.4 Final takeaway

The final takeaway is that credit, duration, liquidity, and safety are related but not interchangeable. A structural three-factor approach lets the paper pull them apart, price them separately, and use them to read market events with much finer economic resolution than a single aggregate bond factor could provide.